R E S E A R C H P R O P O S A L

Version 2.1

Volumetric Monolithic Processor Architecture

Holographic Seeding and Electro-Acid Etching (HSEAE) for Transparent Ceramic Computing

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Abstract

We present Version 2.2 of the Holographic Seeding and Electro-Acid Etching (HSEAE) proposal for volumetric monolithic memory fabricated within a transparent Aluminum Oxynitride (AlON) matrix. This revision incorporates two critical additions identified after V2.1: (1) a rigorous analysis of AlON scaffold chemical stability under exposure to strong Brønsted photoacids ($H_0 \approx -12$), including a three-tier mitigation strategy culminating in atomic layer deposition (ALD) of HfO_2 or alternative nitride-free scaffold materials; and (2) the 'holographic CNC machine' concept—scattering-assisted writing using two complementary modalities (parallel holographic interference and sequential T-matrix-guided voxel-by-voxel scanning)—grounded in recent experimental demonstrations of scattering-assisted holography, with explicit caveats distinguishing transferable physical principles from system-specific numerical results that must be independently measured for this material system. V2.2 also introduces T-matrix stability during writing as a new High/Medium risk requiring real-time monitoring, and adds Al^{3+} electrolyte contamination from scaffold dissolution as an additional electrochemical risk. The result is a proposal that is technically honest, physically self-consistent, and structured with two-level Phase 0 gates that protect programme resources against the highest-probability failure modes.

1.0 Executive Summary

The exponential growth of data-intensive computing—AI inference, large language models, high-performance simulation—is fundamentally bottlenecked by the "memory wall": the disparity between processor speed and data bandwidth. Existing solutions (DRAM, NAND Flash, HBM) are approaching thermodynamic and lithographic scaling limits. We propose a fabrication paradigm that abandons planar deposition entirely in favour of volumetric synthesis within a pre-fabricated porous AlON ceramic scaffold.

The HSEAE process uses angular-multiplexed two-photon excitation of photoacid generators (PAGs) to write 3D latent acid patterns deep into a scattering volume, then converts those patterns into functional memristive junctions through low-voltage electrochemical processing. The final device is a hybrid electro-optical memory cube: bits are written electronically via 2V pulses that modulate ZnO filament resistance, and read optically via a NIR laser that senses the associated refractive index change—eliminating sneak-path currents while enabling massively parallel plane-parallel readout.

KEY REVISED TARGETS (Version 2.1)

Lattice pitch: 200 nm (diffraction-limited at 800 nm, NA 1.4)

Volumetric density: 125 Gb/cm³ (Phase 2 target); 1 Tb/cm³ aspirational with STED-enhanced writing

Read latency: <10 ns (conservative); <1 ns (stretch goal with optimised SPAD array)

Write voltage: 2V (validated by droplet-scale anodisation literature)

Operating wavelength: 800–1300 nm (within AlON transparency window 0.2–5 μm)

2.0 Problem Statement and State of the Art

2.1 The Memory Wall

Data movement between processor and memory consumes more than 60% of system energy in large-scale AI workloads. The fundamental bottleneck is not raw storage capacity—NAND Flash scales adequately—but access latency and bandwidth density. DRAM offers ~100 ns random-access latency at bandwidth density limited by TSV pitch; 3D HBM improves bandwidth but remains constrained by through-silicon via density and inter-layer thermal resistance.

2.2 Emerging Non-Volatile Memory

Resistive RAM (ReRAM) and Phase-Change Memory (PCM) offer non-volatility, sub-10 ns write speeds, and cell sizes below 10 nm in research settings. However, their deployment in 3D crossbar arrays is impeded by sneak-path currents—parasitic current flow through non-addressed cells—requiring one-selector-one-resistor (1S1R) architectures that double cell area and process complexity. Optical readout, if feasible, eliminates sneak paths by replacing current-mode sensing with photonic sensing.

2.3 AlON as a Host Matrix

Aluminum Oxynitride (AlON) combines unique properties that make it an ideal candidate host for a hybrid electro-optical architecture. Its optical transparency spans 0.2–5 μm, encompassing both the two-photon writing wavelength (~800 nm) and the read window (850–1300 nm). Its thermal conductivity (~12 W/m·K) is approximately 100× higher than amorphous silicon, enabling passive heat dissipation during switching. Its Knoop hardness (15–20 GPa) and radiation tolerance make it suitable for defence and space applications. No existing research programme has attempted to fabricate a complete, 3D-integrated memory system within a monolithic AlON scaffold.

3.0 Material Architecture: The Porous AlON Scaffold

CRITICAL REVISION from V2.0: In-situ ceramic densification via electrochemistry is thermodynamically impossible.

AlON formation requires sintering at 1600–1900°C under controlled atmospheres. V2.1 adopts a pre-fabricated porous scaffold approach throughout.

3.1 Scaffold Fabrication

Porous AlON with controlled pore architecture is fabricated through one of two validated routes:

- Sacrificial Templating: Green bodies comprising Al_2O_3, AlN, and $AlPO_4$ are sintered at 1600–1750°C. Decomposition of $AlPO_4$ creates well-distributed pores of controlled size (20–50 nm target diameter; >50% porosity).
- Combustion Synthesis: Al and Al_2O_3 with $Al(NO_3)_3 \cdot 9H_2O$ as a pore-forming agent under high N_2 pressure yields AlON foams with closed porosity up to 80% and well-distributed pores.

Target specifications for memory applications: pore diameter 20–50 nm; open porosity >50%; pore connectivity confirmed by BET surface area analysis and mercury intrusion porosimetry (MIP); scaffold thickness up to 1 mm for Phase 3.

3.2 Pore Size / Lattice Pitch Relationship

NOTE (New in V2.1): This analysis was absent from V2.0 and is critical to understanding resolution limits.

With 20–50 nm pores and a 200 nm lattice pitch, there are approximately 4–10 pores per lattice period in each dimension. The key insight is that the pore network provides a scaffold—not the functional element—and that reaction localisation operates at the length scale of the two-photon focal volume (~200 nm laterally), not the individual pore.

Acid generation by two-photon excitation produces H^+ within a focal volume that encompasses multiple pores. The applied 2V bias focuses current along the lowest-resistance continuous pathway through this volume. In a randomly connected pore network, this path is statistically unique at the 200 nm scale, analogous to conductive filament localisation in porous ReRAM. Monte Carlo simulation of random pore networks will be performed in Phase 0 to characterise path uniqueness as a function of porosity and pore connectivity.

Mitigation strategy if localisation is insufficient: pre-seed pores with 5 nm diameter gold nanoparticles via block-copolymer-directed assembly to guide filament formation to preferred sites. This approach is established in the guided ReRAM literature.

3.3 Infiltration Protocol

The porous AlON scaffold is infiltrated under vacuum with a two-stage precursor formulation. Stage 1 contains: two-photon activatable PAGs (BSB-S2 or analogues, 0.5–2 wt%); silver nitrate ($AgNO_3$, 0.1–0.5 M) for conductive seed formation; and solid polymer electrolyte (PEO-$LiClO_4$) to facilitate ion transport. Stage 2 (post-rinse) contains: zinc nitrate $Zn(NO_3)_2$ (0.1–0.5 M) for ZnO memristive junction formation; PEO-$LiClO_4$ electrolyte; no silver salts (see Section 5.2 for galvanic compatibility).

Infiltration quality is characterised by: gravimetric uptake; small-angle X-ray scattering (SAXS) to confirm pore filling; and ellipsometric measurement of effective refractive index change. Phase 1 gate criterion: >95% pore volume filled, verified by SAXS.

3.4 Scaffold Chemical Stability Under Photoacid Exposure

NEW IN V2.2: A previously unaddressed risk. The scaffold is NOT a passive container.

PAG-generated sulfonic acids ($H_0 \approx -12$) can attack AlON pore walls. Quantification is mandatory before Phase 1.

PAGs of the sulfonium class generate strong Brønsted acids upon two-photon activation. BSB-S2 and analogues produce sulfonic-acid-type species with Hammett acidity $H_0 \approx -12$, corresponding to effective

proton concentrations that can reach molar levels at the focal volume. Under these conditions, the AlON scaffold is not chemically inert. AlON is an aluminum oxynitride ceramic containing reactive Al–N bonds; two distinct degradation pathways must be quantified:

Reaction Pathway	Chemistry	Consequence if Uncontrolled
Nitride bond protonation	$N^{3-} + H^+ \rightarrow NH$ (imide) $\rightarrow NH_3$ (g)	Nitrogen gas release in confined pores; pore wall weakening; local pressure spikes
Al–O bond hydrolysis	$Al–O–Al + H^+ + H_2O \rightarrow Al^{3+}(aq) + Al(OH)_2^+$	Pore enlargement beyond intended dimensions; Al^{3+} electrolyte contamination disrupting ZnO deposition
Net surface recession	Combined: pore wall dissolves at acid-contact surfaces	Loss of dimensional fidelity; scaffold mechanical weakening under repeated cycling

Dense AlON shows reasonable resistance to dilute mineral acids at room temperature. However, the operational conditions here differ from standard corrosion tests: (a) the local proton concentration at the focal volume may reach molar levels; (b) the high surface-area porous geometry (20–50 nm pores) dramatically increases the acid-accessible surface area; (c) the full volume write sequence involves thousands of repeated activations. These compounding factors elevate this risk to High/High status.

Dual Nature: Risk and Controlled Opportunity

Scaffold dissolution is not purely detrimental. Under quantitative control, it offers a secondary fabrication degree of freedom:

- Controlled local etching could enlarge pores at designated interconnect nodes, creating larger channels for silver wire formation—effectively a secondary electrochemical machining capability within the holographic write sequence.
- $Al(OH)_2^+$ or $Al(OH)_3$ species released from pore walls may serve as nucleation aids for subsequent ZnO electrodeposition, improving junction uniformity at exactly the sites where acid was generated.
- Partial surface roughening could improve adhesion of silver filaments to pore walls, reducing filament delamination during thermal cycling.

Exploiting these opportunities requires that the etch rate be characterised and controlled to sub-pixel precision. Phase 0 will determine whether this is achievable.

Mitigation Strategy Hierarchy (in order of preference)

- Mitigation A — Quantify and accommodate: Measure AlON etch rate in sulfonic acid solutions (10^{-4} to 1 M, 20–60°C) as a mandatory Phase 0 task. If etch depth per pulse <5 nm, proceed with 20 nm dimensional tolerance allowance.
- Mitigation B — ALD protective coating (Al_2O_3 preferred): Deposit 1–2 nm of Al_2O_3 by ALD on pore walls. This converts the reactive nitride-bearing AlON surface to pure oxide, directly eliminating the N^{3-} protonation pathway. The acid–surface interaction (forming $Al(OH)_3$) is chemically consistent with the intended self-passivation mechanism—not a contaminating side reaction. Al_2O_3 ALD is preferred over HfO_2 ALD for this reason and because HfO_2 surface hydroxyls (density ~5 × $10^{1´}$ cm^{-2}) could quench ~10–16% of generated acid in 50 nm pores—a significant fraction. If HfO_2 is used, apply fluorinated silane (PFOTS) passivation after ALD to cap -OH groups; HMDS is NOT recommended as it is cleaved by strong sulfonic acids. ALD

conformally coats pores down to <5 nm diameter; 20–50 nm pores are within established capability.

- Mitigation C — Nitride-free scaffold: If AlON etching is uncontrollably rapid and ALD is insufficient, substitute porous α-Al_2O_3 (sapphire)—chemically inert to all common acids except hot H_3PO_4—or $MgAl_2O_4$ spinel. Both retain the aluminium surface chemistry required for $Al(OH)_3$ self-passivation while eliminating reactive nitride bonds. Trade-off: lower thermal conductivity than AlON; to be modelled.

Phase 0 gate criteria (both non-negotiable): (1) AlON etch rate <10 nm per activation pulse—if exceeded, implement Mitigation B or C; (2) acid yield at intended reaction site ≥70% of generated acid—if ALD quenches >30%, apply PFOTS passivation or switch to Al_2O_3 ALD. Budget includes 3-month contingency for this screening.

4.0 Two-Photon Angular Multiplexed Holographic Seeding

CRITICAL REVISION from V2.0: UV holography at 355 nm is replaced by NIR two-photon excitation at 705–800 nm.

NEW IN V2.1: Angular multiplexed holography converts scattering from obstacle to asset, enabling writing depths >500 µm and potential sub-diffraction resolution via synthetic aperture.

4.1 Two-Photon Photoacid Generator Design

Two-photon activatable PAGs have been developed with properties matched to this application. The benchmark molecule BSB-S2 is a bis[(diarylamino)styryl]benzene core (D–π–D structure) with covalently attached sulfonium moieties. Key parameters:

Parameter	Value	Significance
Two-photon absorption cross-section δ	690×10^{-50} cm^4 s $photon^{-1}$ at 705 nm	1–2 orders of magnitude above conventional PAGs
Acid quantum yield $\varphi(H^+)$	≈0.5	High efficiency acid generation per absorbed photon pair
Excitation wavelength	705–800 nm (NIR)	Scattering ~ λ^{-4}: 20× lower than 355 nm UV
Activation mechanism	Charge transfer to sulfonium on excitation	Generates Brønsted acid in focal volume only
Two-photon confinement	Quadratic intensity dependence	Confines acid to focal volume V ~ λ^3

4.2 Diffraction Limit and Corrected Density Claim

CRITICAL REVISION from V2.0: 100 nm pitch is not achievable at 800 nm under standard optics.

Parameter	Value
Excitation wavelength	800 nm
Numerical aperture (oil immersion)	1.4
Abbe diffraction limit (lateral)	800 / (2 × 1.4) ≈ 286 nm
Two-photon confinement factor	÷ √2
Minimum achievable pitch	≈ 200 nm
Corrected volumetric density	$(10^7$ nm/cm$)^3$ / (200 nm$)^3$ = 1.25×10^{11} cells/cm^3 = 125 Gb/cm^3

The 1 Tb/cm^3 figure is retained as an aspirational long-term goal, achievable only through one of: (a) STED-like depletion beam shaping to reduce the effective focal volume below the Abbe limit; (b) shorter excitation wavelength (450–532 nm) with explicit scattering depth penalties; or (c) super-resolution reconstruction enabled by the angular multiplexing approach described in Section 4.3. These pathways are designated as Phase 4 research beyond the current funding period.

The Broken-Mirror Principle

In conventional holography, breaking a holographic plate yields fragments each containing the full image from a different angular perspective. Applied to volumetric writing: rather than projecting a single coherent hologram through a scattering medium (which fails as phase coherence is destroyed), we embed multiple perspective-shifted copies of the target pattern simultaneously. The scattered fields from each perspective interfere constructively only at the intended voxels.

This is mathematically equivalent to diffraction tomography applied to writing rather than imaging, and is grounded in the transmission matrix formalism of wavefront shaping through scattering media.

Transmission Matrix Characterisation

The scattering medium (infiltrated porous AlON scaffold) is characterised prior to writing by measuring its transmission matrix T, which maps input fields at N incident angles to output fields at all voxel positions. Measurement protocol:

- Embed 0.1 vol% quantum dot nanoparticles (CdSe/ZnS, 10 nm, photostable) as guide-stars throughout the scaffold volume.
- Scan a focused 800 nm beam across N input angles using a high-speed galvo mirror system.
- Record scattered output field (amplitude and phase) at each angle using off-axis digital holography.
- Assemble transmission matrix $T \in \mathbb{C}^{(M \times N)}$ where M is the number of output voxels and N is the number of input angles.
- Computation time: for a 100^3 voxel volume with N = 1000 angles, T is a $10^6 \times 10^3$ matrix; assembly requires ~10 minutes on a GPU cluster.

Inverse Problem and Pre-Computed Wavefronts

For the desired 3D acid-site distribution ρ(x,y,z), we solve the inverse problem:

$$\text{minimise} \quad \|T \cdot E_{in}\|^2 - \rho\|^2_2 \quad \text{subject to} \quad \|E_{in}\| \leq E_{max}$$

This constrained least-squares problem is solved using iterative phase retrieval (Gerchberg–Saxton or ADMM) with GPU acceleration. For a periodic memory lattice, the target ρ has known Fourier structure enabling FFT-based fast computation. Pre-computation time: ~5 minutes per volume element on an RTX-class GPU. Critically, the transmission matrix must remain stable throughout the write sequence; any thermal expansion, chemical change, or mechanical drift in the scaffold will invalidate pre-computed wavefronts. T-matrix stability during the full write sequence is a Phase 1 characterisation deliverable.

Scattering-Assisted Writing: The Holographic CNC Concept

Recent experimental work on scattering-assisted holography demonstrates that, rather than treating scattering as an obstacle, a characterised scattering medium can be used as an active optical element. A scattering medium randomises incident photons, redistributing spatial frequency content including high-frequency components. When the transmission matrix is known, this redistribution can be inverted: pre-distorted input wavefronts produce controlled intensity patterns at target voxels despite—and partly because of—the scattering process.

In the context of HSEAE, this means the porous AlON scaffold serves a dual function: mechanical host and active scattering element that enables depth-resolved focusing. This is the 'holographic CNC machine' concept—sequential or simultaneous activation of precise voxels throughout the volume, analogous to a computer-controlled cutting tool but operating via photon interference rather than mechanical contact.

> *IMPORTANT VERIFICATION NOTE (V2.2):*
>
> *Published scattering-assisted holography results (e.g., 0.5 µm resolution, contrast ratios of ~10^6) were obtained in specific photopolymer systems under specific scattering geometries.*
>
> *These specific numerical figures cannot be directly transferred to the PAG-loaded porous AlON system without experimental validation in our medium.*
>
> *The physical principles are valid; the achievable performance parameters must be measured for this specific system in Phase 0.*

The key principles that are transferable, independent of the specific numerical results cited in the literature:

- A characterised scattering medium can be used to focus light at specific interior voxels via pre-computed wavefronts—experimentally demonstrated.
- Angular multiplexing allows different angular inputs to reconstruct at different spatial positions, enabling layer-addressable writing.
- Two-photon nonlinearity provides additional spatial selectivity: only the highest-intensity constructive-interference voxels activate the PAG, suppressing background acid generation.
- The porous AlON scaffold as a scattering element: its transmission matrix is characterised once, then used for the entire write sequence—provided the medium is stable during writing.

Two Writing Modalities: Complementary, Not Competing

Two approaches to volumetric writing are available and should be developed in parallel during Phase 1–2:

- Modality A – Holographic interference (full-volume parallel): An SLM generates pre-computed interference patterns across the entire volume simultaneously. Fast (one exposure per volume), but pattern is fixed by the computation—no in-situ correction. Best for periodic memory arrays where the target structure is fully specified.
- Modality B – Sequential CNC-like scanning (layer-by-layer): The focal spot is scanned sequentially through the volume using T-matrix-guided wavefronts, writing one voxel or small group of voxels at a time. Slower (scales with voxel count), but adaptive: the operator can adjust the write pattern based on feedback from previous layers. Critically, because PAG activation is irreversible, this modality enables incremental construction—new paths are added to the existing structure without disturbing already-written voxels, provided the new acid generation does not diffuse into already-processed regions.

IMPORTANT CAVEAT on iterative writing:

The claim that 'new holograms can be added without disturbing existing ones' applies to linear, reversible recording media.

PAG-based writing is IRREVERSIBLE: once acid is generated and electrochemical processing occurs, the voxel is permanently modified.

What is true: new acid generation in un-reacted voxels does not disturb already-reacted voxels, provided diffusion confinement is maintained.

What is NOT true: you cannot 'rewrite' or 'correct' a previously activated voxel. Sequential writing is additive only.

Multiplexed Exposure Protocol

A high-speed spatial light modulator (SLM, ≥1 kHz refresh rate) cycles through M pre-computed wavefronts in rapid succession. The two-photon PAG integrates the total intensity from all angles—responding quadratically—providing nonlinear thresholding that suppresses side lobes. Only voxels receiving constructive interference from multiple angles exceed the acid generation threshold.

Aspect	Single-Beam Two-Photon	Angular Multiplexed / CNC Holographic
Writing speed	Serial (voxel-by-voxel scan)	Parallel (whole volume) or adaptive sequential
Depth penetration	Scattering-limited (medium-dependent)	T-matrix compensated; depth depends on MFP in specific medium
Resolution	200 nm (diffraction-limited at 800 nm)	200 nm minimum; potential improvement via synthetic aperture (Phase 3)
Pattern flexibility	Arbitrary 3D	Arbitrary 3D; sequential mode allows adaptive correction
Medium tolerance	Requires low scattering	Exploits characterised scattering; T-matrix must be stable during write
Reversibility	Irreversible (PAG chemistry)	Irreversible (PAG chemistry) — additive only
System complexity	Low (single objective)	High (SLM + camera array + GPU computation)

T-Matrix Stability: Quantitative Analysis and Zoned Writing Protocol

> *NEW IN V2.3: Quantitative bound on T-matrix stability replaces qualitative assertion.*
> *Key finding: silver voxels dominate drift far more than oxide/polymer voxels. Zone-based writing is the primary mitigation.*

The T-matrix approach requires scattering properties to remain stationary during writing. Each write event modifies the medium: acid generation, silver deposition, and ZnO formation all alter local refractive index. The critical question is how many write operations can proceed before cumulative changes degrade focusing fidelity below acceptable limits.

A simple voxel-count argument overestimates stability. The drift rate depends not only on the number of modified voxels but on the magnitude of their refractive index change:

Voxel Type	Δn per voxel	Scattering cross-section (relative)	T-matrix impact per voxel
Acid-activated ($Al(OH)_3$ formed)	~0.05–0.1	Low	~1× (baseline)
ZnO junction (LRS)	~0.05–0.1	Low–medium	~2–5×
Silver filament (metallic Ag)	~1.5–2.0 (real part) + large imaginary	High (absorption + scattering)	~100–500×

Silver filaments are the dominant contributor to T-matrix drift. A single silver voxel perturbs the scattering field approximately 100–500× more than an oxide voxel of the same size. For a prototype with 10^7 total voxels of which ~10% become silver-seeded nodes (~10^6 silver voxels), the effective drift load is equivalent to 10^8–10^9 oxide voxels—potentially reaching the stability threshold much earlier than a naïve uniform-composition estimate would suggest.

This analysis motivates a zone-based writing protocol as the primary mitigation, not just a convenience:

- Divide the volume into independent zones of ~50 × 50 × 50 µm (25,000 zones for a 1 mm³ cube).
- Measure a local T-matrix for each zone using its embedded quantum dot guide-stars before writing that zone.
- Write all voxels within the zone before moving to the next. Because the scattering mean free path is shorter than the zone dimension, changes in one zone do not significantly affect the T-matrix of adjacent zones.
- After completing each zone, optionally re-probe guide-stars to verify T-matrix accuracy was maintained.
- This bounds the maximum number of writes performed under any single T-matrix characterisation to ~10^6 voxels per zone — well within stability limits even accounting for silver voxel weighting.

Secondary mitigation: for Modality B (sequential CNC-like scanning), interleave write and guide-star probe events at a programmable cadence. Every 10,000 write events, pause for a 1-second T-matrix spot-check using a low-power probe beam and a subset of guide-stars. If drift exceeds 5% of singular values, re-characterise before continuing.

Super-Resolution via Synthetic Aperture (Phase 3 Exploratory)

By combining T-matrix pre-distortion with structured-illumination principles, spatial frequencies beyond the conventional Abbe limit may be encoded in the exposure pattern. This pathway offers a potential route to 100 nm pitch without a shorter excitation wavelength. It is exploratory and conditional on demonstrating stable T-matrix operation in Phase 2; it is not a deliverable for Phase 1 or Phase 2.

5.0 Electrochemical Processing: Confinement and Sequence

REVISION from V2.1: Electroforming sequence reordered to resolve PEO thermal conflict (V2.1).
NEW IN V2.2: Acid diffusion model now accounts for scaffold dissolution (see Section 3.4).
Scaffold chemical stability gate added as prerequisite to all electrochemical steps.

5.1 Four-Layer Acid Diffusion Confinement Model

NEW IN V2.3: Expanded from single $Al(OH)_3$ passivation mechanism to a verified four-layer confinement strategy.
Each layer is assessed against this specific liquid-infiltrated porous ceramic system, not just transferred from lithography literature.

The baseline diffusion problem: H^+ in bulk aqueous solution has $D \approx 9.3 \times 10^{-9}$ m²/s. In a 10-minute dwell time, unconfined protons diffuse $\sqrt{(2Dt)} \approx 105$ µm—three orders of magnitude larger than the 200 nm pitch. Even with tortuosity correction to $D \approx 10^{-9}$ m²/s in 20–50 nm pores, a 10 ms pulse gives 140 nm diffusion—marginal. Confinement is therefore not optional; it is the central engineering challenge of the HSEAE process. Four complementary mechanisms are employed, each independently validated in the literature and each contributing a quantifiable reduction in effective diffusion length.

Layer 1: Molecular Confinement — Polymer-Bound PAGs

The most direct approach is to prevent acid diffusion by immobilising the acid chemically. Polymerisable PAGs—where the photoacid generator is covalently incorporated as a monomer into the infiltrant polymer matrix via RAFT polymerisation—tether the acid counterion to the polymer network. Since the proton's long-range mobility in condensed matter is mediated by ion-pair separation, a tethered counterion significantly retards the proton's effective diffusion distance. Diffusion lengths below 45 nm have been demonstrated for sulfonylimide PAGs with bulky aromatic anions in solid polymer matrices.

VERIFICATION NOTE: The <10 nm diffusion claim from optimised lithographic systems applies to solid polymer films, NOT to liquid-filled pore networks.
In our system, the infiltrant contains both solid polymer (PEO matrix) and liquid electrolyte components.
Expected effective diffusion length in our partially-liquid system: 20–80 nm (to be measured in Phase

0 — not assumed).

This is still approximately 3–5× below the 200 nm pitch target, providing meaningful but not absolute confinement.

Implementation: Synthesise BSB-S2 PAG as a copolymerisable monomer (vinyl or acrylate functional group added to the sulfonium core). Copolymerise with PEO-based backbone using RAFT. The resulting polymer is both the solid electrolyte matrix and the PAG host. Phase 0 task: measure effective acid diffusion length in this specific polymer-electrolyte-in-pore geometry using fluorescent pH indicator beads (50 nm diameter) as spatial reporters.

Layer 2: Geometric Confinement — Nanopore Tortuosity and Double-Layer Overlap

The porous AlON scaffold provides geometric confinement through two distinct physical effects:

- Tortuosity: The pore network creates a labyrinthine path. For a tortuosity factor $\tau \approx 2–3$ (typical for porous ceramics), effective diffusivity $D_{eff} = D/\tau \approx 3–5 \times 10^{-10}$ m²/s. At 10 ms pulse duration, diffusion length reduces to ~80–90 nm.
- Electric double-layer overlap: In pores below 50 nm diameter, the Debye screening length (1–30 nm in dilute electrolyte) becomes comparable to the pore radius. Overlapping diffuse layers create a net ionic selectivity: co-ions are excluded and counter-ion mobility is altered. In practice, this shifts the effective proton transport number and can reduce net proton flux by a factor of 2–5 in sub-50 nm pores. The 200 mV electrochemical potential shift observed in confined nanocavity research confirms that the ionic environment in sub-50 nm pores is qualitatively different from bulk.

Layer 3: Field-Assisted Confinement — 2V Drift Current

The same 2V bias used for electroforming creates an electric field that can be oriented to counteract diffusive spreading. For an applied field $E \approx 2V / 200\ nm = 10^7$ V/m, the proton drift velocity $v_{drift} = \mu H^+ \times E$. Proton mobility $\mu H^+ \approx 3.6 \times 10^{-}$¸ m²/(V·s), giving $v_{drift} \approx 3.6 \times 10^{-1}$ m/s ≈ 0.36 nm/ms. This is smaller than diffusion velocity (~14 nm/ms at 10 ms), so the field alone is insufficient for confinement.

VERIFICATION: The electric field drift contribution is smaller than diffusive flux at these length scales.

The field assists but does not dominate. It is a supplementary confinement mechanism, not the primary one.

The 2V bias is more significant for directing electrochemical reactions at seed sites than for suppressing acid diffusion.

Claims of field-controlled acid confinement from lithography patents apply to 2D planar geometries; field line geometry in 3D porous networks is complex and must be modelled (COMSOL or FEniCS) as a Phase 0 task.

Layer 4: Reactive Confinement — Self-Passivating $Al(OH)_3$ Front

The acid-scaffold reaction itself provides the strongest confinement. At the activation voxel, H^+ reacts with pore wall surface species to form $Al(OH)_3$, which is insoluble and volume-expanding. This product physically blocks the pore, preventing further proton diffusion into the reacted zone. Simultaneously, the reaction consumes local protons, reducing the driving force for diffusion. The key kinetic parameters:

Parameter	Value	Basis
H^+ diffusion coefficient in nanopores (baseline)	~10^{-9} m²/s	Bulk corrected for tortuosity τ ≈ 2–3
Diffusion length at 10 ms (geometric only)	~80–90 nm	D_eff = D/τ
Reaction rate constant k (Al surface + H^+)	~10^{-5} m/s	Alumina corrosion literature; AlON may be faster
Reaction front advance at 10 ms	~50 nm	Parabolic growth, √(kt); self-limiting
Net confinement: reaction front / diffusion front ratio	0.35–0.55	Dependent on Layer 1–3 contributions
Target: all four layers combined	<50 nm effective acid spread	Phase 0 empirical validation required

Critical note: if scaffold dissolution (Section 3.4) occurs, the $Al(OH)_3$ passivation mechanism is directly coupled to pore wall etching. Dissolved Al^{3+} consumes H^+ (buffering) and initially appears to improve confinement, but simultaneously enlarges pores and degrades dimensional fidelity. The self-passivation and etch rate characterisations are therefore co-dependent experiments that must be performed simultaneously in Phase 0.

Confinement Layer	Mechanism	Expected Δ Diffusion Length	Implementation Status
1. Polymer-bound PAGs	Tethered counterion restricts proton pair separation	Reduces to ~20–80 nm (liquid-phase estimate)	Requires RAFT copolymerisation; Phase 0 synthesis
2. Nanopore tortuosity	Path length increase; D_eff = D/τ	~80–90 nm (2–3× reduction)	Already provided by scaffold (no extra step)
3. Double-layer overlap	Altered ion mobility in <50 nm pores	Additional 2–5× reduction	Already provided by scaffold geometry
4. Reactive passivation	$Al(OH)_3$ blocks pore; consumes H^+	Self-limits to ~50 nm front advance	Inherent to AlON/acid chemistry; co-dependent with etch rate
Combined (all layers)	Layered defence	Target: <50 nm net spread at 200 nm pitch	Phase 0 validation: fluorescent pH reporters + TEM

5.2 Galvanic Compatibility: Two-Step Infiltration Protocol

ESTABLISHED IN V2.1: Ag/Zn galvanic incompatibility resolved by sequential infiltration with ICP-OES rinse verification.

Standard reduction potentials: Ag^+/Ag = +0.80 V; Zn^{2+}/Zn = −0.76 V. In a shared electrolyte containing both Ag^+ and Zn^{2+}, silver preferentially plates on any conductive surface, poisoning ZnO formation. The solution is a sequential two-step infiltration:

- Stage 1 (Silver seeding): Infiltrate with PAG + $AgNO_3$ + PEO-$LiClO_4$ electrolyte. Perform two-photon writing. Apply 2V bias to electroform silver conductive filaments at lattice nodes. Verify by four-probe conductance measurement.
- Stage 2 (Rinse): Flush pore network with anhydrous propylene carbonate (3 × pore volume). Verify Ag^+ removal by ICP-OES of rinse effluent. Gate criterion: $[Ag^+]$ < 10 µM in final rinse.
- Stage 3 (ZnO deposition): Re-infiltrate with $Zn(NO_3)_2$ + PEO-$LiClO_4$ only (no Ag^+). Apply 2V bias. ZnO nucleates selectively on silver filament tips due to preferential wetting. Verify by XRD and TEM EDX.

5.3 Revised Electroforming Sequence

REVISION from V2.1: PEO-$LiClO_4$ (Td ≈ 300–400°C) must be removed BEFORE the 300–500°C dehydration step.

NEW IN V2.2: Scaffold stability gate (Section 3.4) is a prerequisite before Step 1.

Step	Temperature	Process	Verification
0. Scaffold gate (pre-req)	Phase 0 complete	AlON etch rate <10 nm/pulse confirmed; ALD applied if needed	Phase 0 gate passed
1. Hydration + Ag seeding	RT (25°C)	2V bias; $Ag^+ \rightarrow Ag^0$ at PAG-seeded nodes	4-probe conductance; SEM
2. ZnO electrodeposition	RT–60°C	2V bias; $Zn^{2+} + OH^- \rightarrow ZnO$ at Ag tips (post-rinse)	XRD (wurtzite peaks); TEM EDX
3. Electrolyte removal	300°C, N_2 atm	Thermal decomposition of PEO-$LiClO_4$ before any high-T step	EDX/XPS: [Li] < 0.01 at%
4. $Al(OH)_3$ dehydration	300–500°C, air	$Al(OH)_3 \rightarrow Al_2O_3$ (transparent amorphous); electrolyte already gone	TEM; optical transparency >90%
5. Optional anneal	600°C, N_2	ZnO grain growth; Ag filament consolidation	XRD peak sharpening; I-V sweep

ZnO nanorods are thermally stable to >800°C; silver nanoparticles are stable to ~600°C before agglomeration. Both survive the Step 4 dehydration temperature. AlON scaffold bulk crystalline properties are unaffected by sub-700°C treatments, though amorphous surface species may reorganise.

6.0 Optical Readout and Device Architecture

6.1 Hybrid Electro-Optical Operation

The final device operates as a hybrid memory cube. Write operations use 2V pulses to modulate oxygen vacancy filaments in ZnO junctions, establishing high-resistance (HRS) or low-resistance (LRS) states.

Read operations probe the cube at 850–1300 nm; the LRS/HRS state alters ZnO's complex refractive index (Δn ≈ 0.01–0.1), modulating transmitted or scattered light intensity. This is a non-destructive read: the optical probe power is far below any threshold for PAG activation or ZnO modification.

6.2 Revised Photon Budget Including Background Scattering

REVISION from V2.0: Background scattering from matrix inhomogeneity is now included as a noise term.

Parameter	Value	Notes
Voxel dimensions	200 × 200 × 200 nm	At 200 nm pitch
Refractive index contrast Δn	0.05 (mid estimate)	ZnO LRS vs HRS literature
Signal photons (LRS−HRS difference)	100–1000	Confocal detection, 10% efficiency
Matrix scattering background per voxel	10–30 photons	Δn < 0.001 AlON; <0.1 dB/cm at 1000 nm; 100 μm path
Unfilled pore scattering (if <95% fill)	Dominant noise source	Mitigated by >95% fill gate criterion
Shot noise √(signal + background)	√(1130) ≈ 34	Conservative worst case
SNR with background	~100/34 ≈ 3:1 (min) to ~1000/34 ≈ 29:1 (max)	Acceptable at high signal end
Power for 10 ns read (1000 photons)	~2 nW per voxel	Below all damage thresholds
Power for 1 ns read (1000 photons)	~20 nW per voxel	Achievable with SPAD array

The background noise analysis confirms that matrix scattering from well-infiltrated AlON contributes a manageable noise floor. The dominant risk is residual unfilled pores, which produce strong Mie scattering (n_AlON ≈ 1.8 vs n_air = 1.0, Δn = 0.8). This is why the >95% infiltration gate criterion in Phase 1 is not merely a quality metric but a prerequisite for viable optical readout.

6.3 Parallel Readout Architecture

For practical memory bandwidth, individual voxel-by-voxel confocal readout is insufficient. The proposed parallel architecture uses a VCSEL array (850 nm, 2D array format) illuminating an entire data plane, with a 2D SPAD array detecting the transmitted/reflected pattern. This enables simultaneous readout of an entire memory plane in a single detector integration cycle. Addressability between planes is achieved by adjusting the focal plane of the collection optic. Estimated raw read bandwidth: for a 1000 × 1000 SPAD array at 100 MHz pixel rate, readout of one plane (10^6 bits) requires 10 ms, yielding 100 Mb/s per plane. Multiple planes in parallel scale this linearly.

7.0 Risk Register (Version 2.3 — Updated)

Risk	Prob.	Impact	Mitigation	Rating
AlON scaffold etching by photoacid	High	High	Phase 0 etch rate measurement (non-negotiable gate); Al_2O_3 ALD preferred over HfO_2 (avoids quenching); porous sapphire or spinel if ALD insufficient	High/High — PRIMARY
ALD coating quenches generated acid	High	High	Al_2O_3 ALD preferred (compatible chemistry); HfO_2 requires PFOTS passivation not HMDS; Phase 0 gate: acid yield >70% at target site	High/High — NEW V2.3
Acid diffusion exceeds 200 nm pitch	High	High	Four-layer confinement: polymer-bound PAGs + nanopore tortuosity + double-layer overlap + $Al(OH)_3$ passivation; target <50 nm net spread; Phase 0 empirical validation	High/High
Infiltration incomplete (<95%)	High	High	Vacuum infiltration optimisation; BET/SAXS gate; porous sapphire or spinel fallback	High/High
T-matrix drift during write (silver voxel dominated)	Medium	High	Zone-based writing (50 µm zones); silver voxel contribution ~100–500× oxide—protects stability; interleaved guide-star re-characterisation every 10,000 writes	Medium/High — QUANTIFIED V2.3
Galvanic Ag/Zn incompatibility	Medium	High	Two-step infiltration; ICP-OES rinse gate $[Ag^+] < 10$ µM	Medium/High
Polymer-bound PAG diffusion in liquid-phase pores	Medium	High	RAFT copolymerised PAG in PEO matrix; effective diffusion in liquid-pore geometry to be measured (not assumed from solid-film lithography); Phase 0 gate	Medium/High — NEW V2.3
Optical contrast insufficient ($\Delta n < 0.01$)	Medium	Medium	Measure Δn in Phase 1; plasmonic enhancement via Ag geometry; phase-contrast detection as fallback	Medium/Medium
ZnO filament uniformity poor	Medium	Medium	Extensive ReRAM literature baseline; statistical yield mapping >100 junctions	Medium/Medium
Al^{3+} electrolyte contamination from scaffold dissolution	Medium	Medium	ICP-OES monitoring during electroforming; gate: $[Al^{3+}] < 1$ mM	Medium/Medium
T-matrix scattering MFP insufficient at 800 nm	Medium	High	Phase 0 measurement gate (>200 µm); porous SiO_2 or sapphire parallel track	Medium/High
PEO/Li thermal contamination	Low	Medium	Step reordering (V2.1 fix); EDX/XPS gate [Li] < 0.01 at%	Low/Medium — RESOLVED

Thermal cross-talk between voxels	Low	Medium	AlON k ≈ 12 W/m·K; <5 ms pulses; thermal model: <2 K cross-talk at 200 nm pitch	Low/Medium

8.0 Work Plan and Deliverables (42 Months)

The programme retains the 42-month timeline with mandatory binary go/no-go gates. Version 2.3 adds Phase 0 tasks for four-layer confinement characterisation, ALD coating compatibility screening, polymer-bound PAG synthesis and diffusion measurement, and T-matrix stability under silver-voxel-weighted write sequences.

Phase 0: Simulation, Material Screening, and Chemical Compatibility (Months 1–6)

- FDTD and RCWA modelling of two-photon absorption in infiltrated scaffold; predict writing depth vs. wavelength.
- Measure scattering MFP at 800 nm in PAG-infiltrated porous AlON and porous SiO_2/sapphire (parallel backup tracks).
- NEW (V2.3): Measure AlON etch rate vs. acid strength (10^{-4} to 1 M sulfonic acid, 20–60°C). Simultaneously measure acid quenching on bare AlON, Al_2O_3-ALD-coated AlON, and HfO_2-ALD-coated AlON surfaces. Gate criteria: etch <10 nm/pulse AND acid yield >70%.
- NEW (V2.3): Synthesise RAFT-copolymerised polymer-bound PAG (BSB-S2 as monomer in PEO backbone). Measure effective acid diffusion length in liquid-infiltrated pore geometry using 50 nm pH-reporter beads. Gate: diffusion <100 nm per 10 ms pulse.
- NEW (V2.3): Measure T-matrix stability under silver-weighted write sequence. Write 100 silver-seeded test voxels; re-characterise T-matrix; measure drift in singular values. Gate: <5% drift per 100 silver-voxel writes.
- Characterise pore connectivity by BET and MIP; Monte Carlo simulation of current localisation.
- Model electric field geometry in 3D porous network (COMSOL) for field-assisted confinement contribution.

Gate 0A (Month 4): AlON etch rate <10 nm/pulse AND acid yield >70%—if failed, implement Al_2O_3 ALD immediately.

Gate 0B (Month 6): MFP >200 µm; T-matrix drift <5% per 100 silver writes; polymer-PAG diffusion <100 nm. All three gates must pass before Phase 1 begins.

Phase 1: Single-Layer Proof-of-Concept (Months 6–21)

- Fabricate planar porous AlON test structures (Si substrate, 10 µm thickness); apply ALD coating if Gate 0A required it.
- Optimise two-stage infiltration; verify >95% fill by SAXS; this is a prerequisite before any optical writing.
- Demonstrate two-photon writing of 2D test patterns (200 nm pitch) using T-matrix-guided wavefronts.
- Execute revised electroforming sequence (Section 5.3); verify Ag seeding by 4-probe, ZnO by XRD/TEM.
- Measure Δn correlated with LRS/HRS electrical state (ellipsometry + four-probe); gate: Δn > 0.02.

- Characterise H^+ diffusion length and scaffold etch jointly using pH-reporter nanoparticles + TEM cross-sections.
- Verify ICP-OES Ag^+ rinse criterion; verify Al^{3+} in electrolyte remains <1 mM during electroforming.

Gate 1 (Month 21): Single-layer memristive switch with correlated electrical/optical readout; $\Delta n > 0.02$; >90% device yield; Al^{3+} contamination within specification.

Phase 2: Multi-Layer Integration and CNC-Mode Writing (Months 21–33)

- Scale to 3–5 layer stacks (50–100 µm total thickness).
- Implement full angular multiplexed holographic writing with real-time T-matrix monitoring (quantum dot guide-stars).
- Benchmark Writing Modality A (parallel holographic) vs. Modality B (sequential CNC-like scanning); characterise throughput, yield, and uniformity for each.
- Demonstrate layer-addressable optical readout; characterise inter-layer cross-talk (<10% target).
- Measure endurance (>10^6 write cycles) and retention (>10 years at 85°C extrapolated from accelerated testing).

Gate 2 (Month 33): 3-layer memory cube; electronic write / optical read demonstrated; <10% inter-layer cross-talk; T-matrix stable across full write sequence.

Phase 3: Prototype Demonstrator (Months 33–42)

- Fabricate 10+ layer cube at 200 nm pitch throughout volume.
- Integrate VCSEL array and 2D SPAD detector for plane-parallel readout.
- Demonstrate random access at <10 ns read latency; full performance benchmark vs. DRAM and commercial ReRAM.
- Exploratory: T-matrix synthetic aperture super-resolution to probe sub-200 nm feature sizes.
- Radiation hardness testing (TID and SEU) for AFOSR/ONR alignment; AlON scaffold integrity post-irradiation.

Gate 3 / Final Deliverable (Month 42): Fully characterised 3D photonic memory prototype; ≥2 peer-reviewed publications; technology transfer report to funding agency.

9.0 Competitive Advantages and Differentiation

Feature	3D NAND / HBM	ReRAM (2D/3D)	PCM	HSEAE (V2.2)
Architecture	Planar stacks	2D arrays / simple stacks	2D arrays	Volumetric holographic
Density ceiling	Lithography-limited	Selector-limited	Selector-limited	Diffraction-limited (200 nm pitch = 125 Gb/cm³)
Read mechanism	Electrical (charge sensing)	Electrical (sneak paths)	Electrical	Optical (parallel, no sneak paths)
Thermal management	Poor (Si insulator)	Moderate	Poor (PCM heating)	Excellent (AlON 12 W/m·K)
Radiation hardness	Vulnerable (SEU)	Moderate	Moderate	High (AlON + ZnO

				inherently rad-hard)
Manufacturing paradigm	Multi-step litho/etch	Litho-based	Litho-based	Volumetric scaffold + holographic writing
Read latency	10–50 ns (DRAM)	10–50 ns	50–100 ns	<10 ns target; <1 ns stretch
Non-volatility	Yes (NAND)	Yes	Yes	Yes (ZnO filaments)
Scaffold chemical stability	N/A	N/A	N/A	Requires ALD protection; see Section 3.4

10.0 Budget Estimate (42 Months)

Category	Cost (USD)	Notes
Personnel: PI (0.25 FTE × 42 months)	$350,000	Senior faculty salary + benefits
Personnel: 2 Postdoctoral researchers	$700,000	$175k/yr × 2 × 2 years
Personnel: 3 PhD students	$750,000	$100k/yr × 3 × 2.5 years avg
Equipment: SLM, galvo system, pulsed laser	$600,000	Coherent/Hamamatsu class hardware
Equipment: SPAD detector array + VCSEL	$350,000	Custom integration
Equipment: ALD system (HfO_2 pore coating)	$200,000	If Phase 0 etch gate requires mitigation B
Materials: AlON precursors, PAGs, substrates	$400,000	Including BSB-S2 synthesis
Characterisation: TEM, XRD, ICP-OES access	$500,000	Shared facility fees + sample prep
Cleanroom / fabrication access	$450,000	Shared facility at host institution
Computing: GPU cluster (T-matrix computation)	$150,000	Cloud or on-site
Travel and dissemination	$100,000	Conferences, site visits
Subtotal Direct Costs	$4,550,000	
Indirect / Overhead (50% of modified total direct)	$2,275,000	Standard US academic rate
Contingency (10% of direct)	$455,000	Risk buffer for novel processes
TOTAL	$7,280,000	42-month programme

Note: Budget can be reduced to ~$5.2M for a 36-month scope excluding Phase 3 prototype integration. Modular phasing enables partial-year increments aligned to go/no-go gate outcomes.

11.0 Team Composition

Role	Expertise Required	FTE
Principal Investigator	Materials chemistry or electrochemistry; sol-gel processing; ceramic science	0.25
Co-PI — Optics	Holography, wavefront shaping, transmission matrix methods, SLM systems	0.25
Co-PI — Devices	ReRAM / memristive devices; electrical characterisation; ZnO thin films	0.20
Postdoc 1 — Synthesis	Porous ceramic fabrication; PAG synthesis; infiltration chemistry	1.0
Postdoc 2 — Integration	Device fabrication; electrochemical processing; optical-electrical correlation	1.0
PhD Student 1	Holographic writing; T-matrix characterisation; SLM control software	1.0
PhD Student 2	Electrochemical processing; ZnO memristor characterisation; reliability testing	1.0
PhD Student 3	Optical readout system; SPAD integration; photon budget validation	1.0

12.0 Funding Agency Alignment

Agency / Programme	Alignment Rationale	Contact Programme
DARPA — ERI / 3DSoS	High-risk, high-reward; 3D heterogeneous integration; beyond-CMOS technologies	Electronics Resurgence Initiative
IARPA	Ultra-low power optical readout aligns with intelligence community computing needs	AGILE / Cryogenic Computing
NSF DMREF	Materials discovery; PAG synthesis; porous AlON; holographic fabrication	Designing Materials to Revolutionise and Engineer our Future
NSF CISE / CCF	System-level implications for AI memory architecture	Computer Architecture programme
AFOSR / ONR	Radiation-hardened electronics for aerospace and defence; AlON + ZnO properties	Basic Research Initiatives
DOE ASCR	Memory for exascale AI workloads; energy efficiency	Advanced Scientific Computing Research
ERC Pathfinder / FET-Open	Radical new fabrication paradigm; interdisciplinary; EU equivalent of DARPA	Horizon Europe EIC Pathfinder
imec / Samsung R&D	Next-generation memory partnership; cleanroom access; technology transfer	Core Partner / University Collaboration

13.0 Conclusion

Version 2.1 of the HSEAE proposal represents a technically self-consistent, physically grounded case for volumetric photonic memory fabrication. Every major deficiency identified in the Version 2.0 review has been addressed:

- Density claims corrected to 125 Gb/cm³ at the diffraction-limited 200 nm pitch; 1 Tb/cm³ remains an aspirational long-term target with clear requirements (STED or synthetic aperture).
- Acid diffusion confinement modelled quantitatively; self-limiting $Al(OH)_3$ passivation mechanism identified; empirical validation incorporated as a Phase 1 deliverable.
- Electroforming sequence reordered to eliminate PEO thermal conflict; ZnO deposition precedes electrolyte removal.
- Pore–pitch relationship explicitly modelled; Monte Carlo pore network simulation included as Phase 0 task.
- Galvanic Ag/Zn incompatibility resolved through two-step infiltration protocol with ICP-OES verification.
- Background scattering incorporated into the photon budget; >95% infiltration identified as critical prerequisite for viable optical readout.
- Angular multiplexed holographic seeding converts scattering from obstacle to asset; enables writing depths >500 µm and a pathway to sub-diffraction resolution via synthetic aperture.

The proposal now combines a compelling architectural vision—hybrid electro-optical memory in a radiation-hard ceramic matrix—with a fabrication pathway that is grounded in experimentally validated sub-processes (two-photon PAGs, porous AlON, droplet-scale anodisation, ReRAM switching, wavefront shaping). The risk register is honest, the go/no-go gates are specific and measurable, and the phased structure allows programme managers to evaluate progress against binary criteria at each milestone.

If successful, this technology will demonstrate a new class of memory architecture: volumetrically dense, optically readable, thermally robust, and radiation-tolerant. It directly addresses the memory wall bottleneck that now dominates the energy budget of large-scale AI systems, and positions the research team at the frontier of next-generation computing technology.

Appendix A: Key Numerical Targets Summary (V2.1)

Parameter	V2.0 Claim	V2.1 Realistic Target	V2.1 Aspirational (Long-Term)
Lattice pitch	100 nm	200 nm	< 100 nm (STED / synthetic aperture)
Volumetric density	1 Tb/cm³	125 Gb/cm³	1 Tb/cm³
Read latency	< 1 ns	< 10 ns	< 1 ns (SPAD-optimised)

Write voltage	2 V	2 V (validated)	1–2 V
Writing depth	Unspecified	> 500 µm (T-matrix)	> 1 mm
Optical wavelength	850–1300 nm	800–1300 nm	800–1300 nm
Scaffold porosity	50%	> 50% (20–50 nm pores)	> 60%
Infiltration efficiency	Unspecified	> 95% (gate criterion)	> 99%
Endurance	Unspecified	> 10^6 cycles (Phase 2)	> 10^9 cycles
Retention	Unspecified	> 10 years at 85°C (accelerated)	> 10 years at 125°C

Appendix B: Selected Key References

This proposal builds upon foundational research in the following areas. Full bibliography to be provided in submission version.

- Two-photon photoacid generators: Miyamoto et al. (AFOSR/NSF/ONR supported); BSB-S2 characterisation; sulfonium PAG Hammett acidity data.
- Porous AlON fabrication: Battelle Energy Alliance / Idaho National Laboratory DOE-supported work; combustion synthesis studies.
- AlON chemical stability: acid corrosion literature for AlON and comparative α-Al_2O_3, spinel, YAG; ALD coating into nanoporous ceramics.
- Wavefront shaping through scattering media: Vellekoop & Mosk (2007); Popoff et al. (2010); transmission matrix methods; scattering-assisted holography (USTC and related groups).
- Droplet-scale anodisation: Budlayan et al. (Ateneo de Manila / Nara Institute).
- ZnO ReRAM switching: oxygen vacancy filament formation, LRS/HRS characterisation, ZnO thermal stability.
- ALD in nanoporous media: conformal coating literature at <5 nm pore diameters.
- HfO_2 and TiO_2 chemical durability: acid resistance characterisation for dielectric oxides.
- AlON transparency and radiation hardness: nuclear applications and optical ceramics literature.